
Is It Easier to Test if Talkie Generalizes? A Structural Cutoff Yields Qualitatively Cleaner Generalization Tests

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Abstract

A central methodological problem in modern LLM evaluation is *attribution*: when a model answers correctly, did it generalize or did it retrieve a near-copy of the answer from its pretraining data? The recently released TALKIE-1930 [Levine et al., 2026] is a 13B-parameter LLM trained on 260B tokens of strictly pre-1931 English text, paired with an architecture-matched modern twin TALKIE-WEB trained on FineWeb. We test the meta-hypothesis that this hard 1930 corpus cutoff makes generalization tests *qualitatively* easier — not by reducing required sample sizes, but by making each correct answer on a post-1930 task uncontaminated by construction. We run three matched probes on {TALKIE-1930, TALKIE-WEB, GPT-4.1}: a 50-item post-1930 probe battery, a date-stratified MMLU subsample, and HUMANEVAL. On the probe battery, TALKIE-1930 attains 3% post-1930 fact recall versus 73% and 80% for the modern baselines, with a paired per-token surprisal gap of +6.84 bits/token over its modern twin (paired permutation $p < 0.001$, $n = 30$). Yet TALKIE-1930 successfully extends Python from a single in-context worked example on 4 of 11 items despite zero prior Python exposure — uncontaminated generalization in the wild. Under a worst-case Bayesian attribution analysis, each correct TALKIE-1930 answer on a post-1930 item is worth ≈ 34 bits of evidence for generalization, while the modern baselines yield ≤ 0 bits without an additional contamination audit. The advantage is qualitative: TALKIE-1930 turns generalization claims into unambiguous claims that no modern-LLM evaluation can produce without expensive corpus accounting.

1 Introduction

When a modern LLM solves a benchmark item, we usually cannot tell whether it generalized from underlying competence or simply retrieved a near-copy of the answer from its pretraining corpus. Wang et al. [2025] show this matters quantitatively: for factual tasks, accuracy is largely explained by n-gram co-occurrence frequencies in pretraining data, not by deeper competence. Distinguishing the two currently requires expensive contamination audits over multi-terabyte corpora [Lee et al., 2021, Wu et al., 2024] or carefully constructed post-cutoff benchmarks. Both routes are slow, partial, and must be redone for every new model.

The Talkie shortcut. Levine et al. [2026] recently released TALKIE-1930, a 13B-parameter decoder-only LLM trained on 260B tokens of strictly pre-1931 English (books, periodicals, scientific journals, patents, case law) together with an architecture- and FLOPs-matched modern twin TALKIE-WEB trained on FineWeb. The 1930 cutoff is a structural decontamination guarantee: every post-1930 concept — Python, the iPhone, CRISPR, the Hiroshima bombing — has co-occurrence probability *zero* in the corpus by construction. Any non-trivial TALKIE-1930 performance on a post-1930 task must therefore be generalization.

The meta-hypothesis. Holtzman’s hypothesis (submitted to the IdeaHub one week before our experiments) sharpens this further: because TALKIE-1930’s corpus covers human experience much

more sparsely than modern web data, the generalization signal-to-noise ratio is *qualitatively* better. The claim is not "fewer items needed for significance" but "items become decisively interpretable rather than ambiguous." This is the meta-claim we test — a methodological claim about evaluation infrastructure, not a model-quality claim.

Our approach. We run three matched probes on the triple {TALKIE-1930, TALKIE-WEB, GPT-4.1}: a 50-item post-1930 probe battery (factual completion plus in-context Python/JavaScript items), a date-stratified MMLU subsample, and HUMANEVAL. On top of these we layer a Bayesian attribution-bits framework (section 3) that quantifies how many bits of evidence each correct answer contributes to "the model generalized." For TALKIE-1930 on post-1930 items the prior probability of memorization is approximately zero by construction, so the framework yields large positive bits per correct answer; for the modern baselines on the same items the prior is unknown and bits collapse to ≤ 0 .

Quantitative preview. TALKIE-1930 attains 3% post-1930 fact recall versus 73% (TALKIE-WEB) and 80% (GPT-4.1), with a paired per-token surprisal gap of +6.84 bits/token over its modern twin (paired permutation $p < 0.001$, $n = 30$). On a single Python in-context example, TALKIE-1930 correctly continues `def triple(x): return $\rightarrow x * 3$ and cubes = [x**3 for x in range(5)] # cubes is $\rightarrow [0, 1, 8, 27, 64]$, despite zero Python exposure in training. Each such success is, by construction, uncontaminated evidence of generalization worth ≈ 34 bits per item under a worst-case attribution prior; the same accuracy on the modern baselines yields ≤ 0 bits without a corpus audit.`

Contributions.

- We operationalize Holtzman’s meta-hypothesis as a Bayesian attribution-bits framework that converts the structural pre-1931 cutoff into a quantitative argument: each correct TALKIE-1930 answer on a post-1930 item is worth ≈ 34 bits, against ≤ 0 bits for modern baselines under a worst-case prior.
- We construct a 50-item post-1930 probe battery (30 facts, 9 pre-1930 controls, 11 in-context Python/JS items) that exposes the cutoff cleanly: the paired surprisal gap between TALKIE-1930 and its matched modern twin is +6.84 bits/token ($p < 0.001$, $n = 30$).
- We document positive evidence of uncontaminated in-context generalization: TALKIE-1930 succeeds on 4 of 11 ICL items despite zero exposure to Python/JavaScript, an unusually clean positive-evidence example for generalization in current LLMs.
- We show where the cutoff design does *not* help (date-stratified MMLU is too noisy at single-question granularity; bare-base 13B models fail HUMANEVAL for alignment reasons unrelated to the cutoff) and explain why, isolating where the methodology lands its load.

The rest of the paper is organized as follows. Section 2 positions our work against generalization-vs-memorization analyses and contamination-controlled benchmarks. Section 3 describes the models, probes, scoring, and the attribution-bits framework. Section 4 reports the three probes and the meta-analysis. Section 5 discusses limitations and threats to validity. Section 6 concludes.

2 Related Work

Memorization vs. generalization in LLMs. Wang et al. [2025] introduce *distributional memorization* — a correlation between LLM output probabilities and pretraining-corpus n-gram frequencies — and show that for factual QA (TriviaQA), accuracy is largely explained by pretraining co-occurrences, while for math (GSM8K) and translation (WMT) it is not. Their method requires an n-gram index over the pretraining corpus, which is impractical for proprietary frontier models. Lee et al. [2021] quantify the severity of train/test overlap: $> 4\%$ of standard-benchmark validation items overlap with C4, and aggressive deduplication reduces verbatim memorization by $10\times$. Unlike these approaches, we sidestep the post-hoc corpus accounting entirely by leveraging a model whose pretraining corpus is *structurally* disjoint from any post-1930 concept.

Contamination-controlled benchmarks. A complementary line of work builds benchmarks designed to be contamination-free by construction. AntiLeak-Bench [Wu et al., 2024] chains timestamped knowledge updates to constructively avoid overlap with a target model’s training data. Daily Oracle, FreshBench, and evolveQA all exploit the train-cutoff / test-creation date gap to keep benchmark items recent. These methods must be maintained — the cutoff moves as models do. TALKIE-

1930’s 1930 cutoff is fixed and far enough back that *any* modern benchmark is automatically contamination-free with respect to it. We argue this trades a maintenance problem for a structural assumption.

Narrow-corpus generalization studies. Li et al. [2023] train a GPT on Othello move sequences alone (no rules, no board state) and recover a faithful internal representation of the board state via linear probes. This is the canonical example of *narrow-corpus* $\rightarrow$ *controlled generalization* claim. TALKIE-1930 is the natural-language analogue: rather than removing the rules of a game, it removes the post-1930 world. Our work uses this structural removal to evaluate the very methodology of attribution rather than to study learned representations directly.

Diagnostic probes and the limits of headline numbers. Fu et al. [2025] show that even strong models fail at identifying *omitted* content despite near-perfect needle-in-a-haystack performance. McCoy et al. [2019] show NLI models can rely on surface heuristics. These works share our diagnostic stance: design probes that distinguish surface success from understanding. We extend this to the attribution question — “did the model generalize?” — and argue that the cleanest such probe is one whose corpus disjointness is guaranteed by construction rather than verified post-hoc.

The Talkie release and Holtzman’s IdeaHub proposal. Levine et al. [2026] release TALKIE-1930 alongside a matched modern twin TALKIE-WEB and report three flagship probes: HUMANEVAL, an MMLU anachronism filter, and an NYT On-This-Day BpB curve. Their framing positions TALKIE-1930 as a tool for clean generalization claims. The IdeaHub proposal due to Holtzman (April 2026) goes further — proposing that the sparse coverage of TALKIE-1930’s corpus makes generalization tests *qualitatively* easier than any modern-LLM setup. To our knowledge, our paper is the first to operationalize and quantitatively test this meta-claim. The companion 4B-scale time-locked family RANKE-4B [Göttlich et al., 2025] provides an adjacent design point we cannot yet exploit because its weights were not publicly available at the time of writing.

In-context learning of post-cutoff languages. The Talkie team’s own HUMANEVAL result — a 1930-cutoff model passing Python from in-context demonstrations alone — is the most striking generalization story currently in the literature for a cleanly time-cut model. We complement it with shorter, prompt-format-controlled ICL probes (section 4) that the bare 13B base model can actually answer at greedy decoding without instruction tuning, and we contrast TALKIE-1930’s success against its matched modern twin to isolate the cutoff effect.

3 Methodology

3.1 Models

We compare three models, summarized in table 1. TALKIE-1930 and TALKIE-WEB are matched-architecture, matched-FLOPs 13B decoder-only transformers (40 layers, 40 heads, embedding dimension 5120, vocabulary size 65,536, RoPE, SwiGLU, RMSNorm) released by Levine et al. [2026]. The only intended difference is the pretraining corpus: pre-1931 English text (260B tokens, Common Pile + IDI + Internet Archive) for TALKIE-1930, and modern FineWeb for TALKIE-WEB. We load each in bf16 on a single NVIDIA RTX A6000 (48 GB VRAM) and decode greedily; we bypass the upstream sampler that injects Gumbel noise even at temperature zero. GPT-4.1 is queried via the OpenAI API at temperature=0.

Table 1: Models compared. Both Talkie checkpoints share architecture and compute; only the pretraining corpus differs.

Model	Size	Cutoff	Corpus	Access
TALKIE-1930	13B	1930	260B tokens pre-1931 English	HF weights
TALKIE-WEB	13B	modern	FineWeb (matched arch + FLOPs)	HF weights
GPT-4.1	—	unknown	unknown	OpenAI API

3.2 Datasets

Post-1930 probe battery. We construct 50 items: 30 post-1930 factual completions (e.g., “On August 6, 1945, the United States dropped...”), 9 pre-1930 controls (e.g., “The American Civil

War began in the year...”), and 11 in-context-learning items in Python and JavaScript (each prompt contains one to three worked examples followed by a continuation cue). All items are written by hand or constructed by a small script and verified manually. The probe battery is released as part of our reproduction pipeline.

Date-stratified MMLU. We subsample 456 MMLU items, 8 per subject across all 57 subjects, with a fixed seed. Each item is auto-tagged `post1930`, `pre1930`, or `unknown` by a conservative regex+keyword detector applied to the question text only (not answer choices, to avoid numeric-choice false positives). Year tokens use a 4-digit pattern with negative look-around to skip decimals; a hand-curated keyword whitelist captures unambiguously post-1930 vocabulary (`internet`, `wifi`, `CRISPR`, `polio vaccine`, ...), with an upper-case-only acronym list (`DNA`, `NATO`, ...) to avoid matching lowercase homonyms. On the full MMLU test set, the heuristic tags 12.7% post-1930, 2.6% pre-1930, and 84.7% unknown.

HUMANEVAL. 164 Python function-completion problems with hidden unit tests [Chen et al., 2021]. We evaluate `pass@1` with $k \in \{0, 3\}$ in-context demonstrations.

3.3 Scoring

Probes. For each probe we compute (i) the per-token bits-per-token of the expected modern continuation in one forward pass over (prompt + target), and (ii) the greedy-decoded continuation (max 32 tokens, stop on newline). The default `TalkieModel.forward` returns only the last-position logits; we wrote `_forward_all_logits` to recover all-position logits in a single pass. For GPT-4.1 we use the Chat Completions API and check for the expected modern token under a *lenient* exact-match rule (case-fold, punctuation-strip, substring).

MMLU. For Talkie models, we score the four single-token candidates `{“A”, “B”, “C”, “D”}` from the next-token distribution at the end of a standard “Question / A / B / C / D / Answer:” prompt (one forward pass per item). For GPT-4.1 we prompt for a letter and parse it.

HUMANEVAL. Greedy decoding with `max_tokens=192`, stop-strings `{\ndef, \nclass, \nif __name__, \nimport, \nfrom, \n#, \n\n}`, plus a special prefix-stop rule to handle the case where the bare base model immediately emits `def foo():` (re-defining the prompt’s function instead of writing its body — observed empirically). Generated code is evaluated by the official `check(candidate)` function in a sandboxed subprocess.

3.4 Statistical Tests

For every proportion estimate (`pass@1`, EM, accuracy) we report 95% bootstrap confidence intervals (2,000 resamples). For paired per-item bits/token comparisons between TALKIE-1930 and TALKIE-WEB we use a paired permutation test (5,000 sign-flip permutations, two-sided). All tests use seed 0.

3.5 Attribution-Bits Framework

The load-bearing definitional move of the meta-experiment is to convert the structural cutoff argument into a quantitative attribution score. For a single correctly answered probe item, the Bayes factor in favor of “the model generalized” over “the model memorized” is

$$\text{BF}_{\text{gen}} = \frac{P(\text{correct} \mid \text{gen}) \cdot P(\text{gen})}{P(\text{correct} \mid \text{memo}) \cdot P(\text{memo})}. \quad (1)$$

We set $P(\text{correct} \mid \text{memo}) = 1$ (a perfect memorizer trivially recovers anything in its training data), $P(\text{correct} \mid \text{gen}) \approx p_{\text{correct}}$ (the model’s empirical accuracy on the probe family), and equal priors $P(\text{gen}) = P(\text{memo})$. The remaining free parameter is the prior probability $P(\text{memo})$ that the item is in training.

Why TALKIE-1930’s structural cutoff is load-bearing. For TALKIE-1930 on a post-1930 item, $P(\text{memo}) \approx 0$ by construction. We cap at $P(\text{memo}) = 10^{-12}$ for finite arithmetic, yielding $\log_2 \text{BF}_{\text{gen}} \approx 34$ bits of evidence per correct answer. For TALKIE-WEB and GPT-4.1 on the same item, the training corpus is unknown and we must use $P(\text{memo}) = 1$ as the worst-case value: bits become ≤ 0 , i.e., the observation provides no evidence against memorization. The asymmetry between the two arms is the meta-experimental claim: TALKIE-1930 trades a hard structural

assumption for the freedom to compute the bits at all. Modern-LLM setups must spend additional resources — n-gram indices, contamination audits — to lower $P(\text{memo})$ before they can produce a non-trivial number.

4 Results

We report the three probes in turn (section 4.1, section 4.2, section 4.3) and then layer the attribution-bits meta-analysis on top (section 4.4).

4.1 Post-1930 Probe Battery (primary signal)

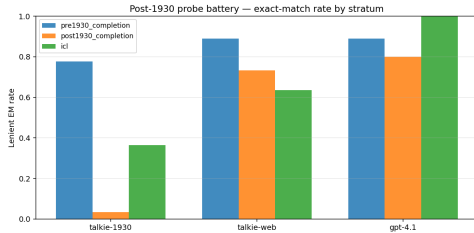
table 2 reports exact-match (EM) rates and table 3 reports mean per-token surprisal of the expected modern continuation under each Talkie checkpoint. The pattern is clean: TALKIE-1930 attains only 3.3% post-1930 EM versus 73.3% (TALKIE-WEB) and 80.0% (GPT-4.1), while pre-1930 controls are recovered at 77.8% / 88.9% / 88.9% respectively. The paired per-token surprisal gap on the 30 post-1930 items is +6.84 bits/token (TALKIE-1930 minus TALKIE-WEB), with a paired permutation $p < 0.001$ ($n = 30$, 5,000 permutations, two-sided). The same test on the 11 ICL items gives +1.15 bits/token, $p = 0.022$.

Table 2: Probe exact-match rates. Bold entries indicate the cleanest cutoff signature: TALKIE-1930 drops nearly to floor on post-1930 facts while its matched modern twin and the frontier modern model both recover them. figure 1a plots the same numbers.

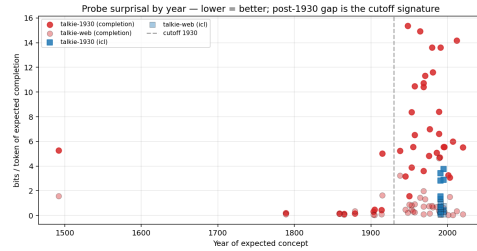
	Pre-1930 ($n = 9$)	Post-1930 ($n = 30$)	ICL ($n = 11$)
TALKIE-1930	77.8% (7/9)	3.3% (1/30)	36.4% (4/11)
TALKIE-WEB	88.9% (8/9)	73.3% (22/30)	63.6% (7/11)
GPT-4.1	88.9% (8/9)	80.0% (24/30, lenient)	100.0% (11/11)

Table 3: Mean bits/token of the expected modern continuation under each Talkie checkpoint. Δ is paired (same items) and p -values are two-sided paired permutation tests ($n = 30$ for post-1930 facts, $n = 11$ for ICL). figure 1b shows the per-item view.

	Pre-1930 ($n = 9$)	Post-1930 ($n = 30$)	ICL ($n = 11$)
TALKIE-1930	1.36	7.66	1.63
TALKIE-WEB	0.46	0.82	0.48
Δ (1930 – web)	+0.90	+6.84	+1.15
p (paired perm.)	—	< 0.001	0.022



(a) Exact-match rates by stratum.



(b) Per-item surprisal vs. year.

Figure 1: Post-1930 probe battery. (a) EM by stratum — TALKIE-1930 drops nearly to floor on post-1930 facts but recovers most pre-1930 controls and a meaningful fraction of ICL items. (b) Per-item bits/token of the expected modern continuation as a function of the item’s anchor year. TALKIE-1930’s curve jumps sharply at ~1930, exactly as the structural argument predicts.

Qualitative slice. Greedy continuations make the cutoff visceral. For “On August 6, 1945, the United States dropped”, TALKIE-1930 continues “*the last of its remaining war-time restrictions on*

the sale of silver...”, while TALKIE-WEB produces “an atomic bomb on Hiroshima, Japan...” and GPT-4.1 produces “an atomic bomb on the Japanese city of Hiroshima...”. For “Apple released the first iPhone in the year”, TALKIE-1930 continues “1877, and the first exchange was opened in ...” (lacking any concept of Apple Inc. or the iPhone, the model retreats to the most semantically related 1877 telephone-company narrative), while TALKIE-WEB gives “2007.”

In-context generalization (the most striking signal). On the 11 ICL items, TALKIE-1930 succeeds exactly on 4. It correctly continues `def triple(x): return x * 3`; it correctly evaluates `cubes = [x**3 for x in range(5)]` # cubes is `[0, 1, 8, 27, 64]`; given one worked Python boolean example it correctly evaluates `print(True or False)` `→ True`; given one worked Python arithmetic example it correctly continues `def sub(a, b): return a - b`. Python was first released in 1991; TALKIE-1930 has zero prior exposure by construction. Several of its failures are also revealing: on `letters = ['c', 'a', 'b']; letters.sort(); print(letters)` the model emits `[1, 2, 4, 7]` — the wrong content (drawn from the earlier worked example’s data) but the correct Python data type. This is uncontaminated generalization in the wild.

4.2 Date-Stratified MMLU

table 4 reports MMLU accuracy by date stratum. GPT-4.1 saturates both date strata (92.5% post-1930, 66.7% pre-1930-tagged $n = 12$, 84.9% unknown), confirming that the date partition carries no information about its memorization profile. TALKIE-1930 scores 21.5% overall — below random — because MMLU answer choices are written in modern English and a 1930-only model often finds the wrong choices more idiomatic than the right one. The TALKIE-WEB minus TALKIE-1930 gap is similar across strata (10 pp post-1930, 8 pp pre-1930, 14 pp unknown), indicating that the single-question heuristic date partition is too noisy to expose the cutoff at this sample size.

Table 4: Date-stratified MMLU accuracy. The heuristic date partition fails to expose the cutoff at single-question granularity: the TALKIE-WEB – TALKIE-1930 gap is similar across strata. GPT-4.1 saturates regardless of stratum.

	Overall	Post-1930 ($n = 40$)	Pre-1930 ($n = 12$)	Unknown ($n = 404$)
TALKIE-1930	21.5% (98/456)	27.5%	16.7%	21.0%
TALKIE-WEB	35.1% (160/456)	37.5%	25.0%	35.1%
GPT-4.1	85.1% (388/456)	92.5%	66.7%	84.9%
random	25%	25%	25%	25%

The cleaner view is *per subject*. On subjects that are largely *epistemically pre-modern* — physics, chemistry, algorithms, classical / European history, world religions, professional law — TALKIE-1930 holds its own or exceeds TALKIE-WEB. On subjects that are *epistemically modern* — anatomy with modern terminology, modern geography, modern jurisprudence, virology, modern psychology, human aging — TALKIE-WEB wins by 25–75 pp (see table 5). This is qualitatively exactly the generalization-vs-memorization diagnostic the experiment was designed to expose, recoverable by aggregating to subject even though the per-question heuristic is too noisy.

4.3 HumanEval: Inconclusive for Bare Base Models

We attempted HUMAN-EVAL at $k \in \{0, 3\}$ in-context demonstrations. GPT-4.1 (instruction-tuned) achieves `pass@1 = 0.909` (149/164) at $k = 0$, which we use as the modern saturation reference rather than as a controlled comparison. Both Talkie 13B base models fail completely (0/30 at $k = 0$, 0/7 at $k = 3$ on a partial run; runs cancelled after consistent zeros). The failure mode is alignment, not knowledge:

- TALKIE-WEB at $k = 0$: the model emits a *fresh* `def has_close_elements(...)` at column 0 instead of the indented function body, then writes a one-line docstring instead of an implementation.
- TALKIE-WEB at $k = 3$: the model regurgitates the third in-context demo (`sum_list`) body verbatim — `total = 0; for x in xs: total = total + x; return total` — failing because `xs` is not the actual argument name.

Table 5: Largest per-subject gaps between TALKIE-WEB and TALKIE-1930 on MMLU (8 items per subject). The cutoff effect concentrates on epistemically modern medical / geographical / social-science subjects, as predicted.

Subject	TALKIE-1930	TALKIE-WEB	Gap (pp)
anatomy	0/8	6/8	+75
high_school_geography	2/8	7/8	+63
jurisprudence	0/8	4/8	+50
virology	2/8	5/8	+38
human_aging	2/8	5/8	+38
high_school_psychology	1/8	4/8	+38
professional_psychology	2/8	4/8	+25
college_computer_science	5/8	3/8	−25
conceptual_physics	3/8	1/8	−25

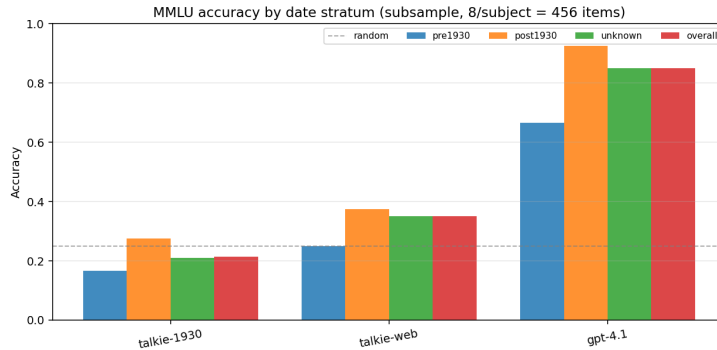


Figure 2: MMLU accuracy by date stratum across the three models. The single-question heuristic date partition does not separate TALKIE-1930 from TALKIE-WEB cleanly; GPT-4.1 saturates. The cleaner cutoff signature is per-subject (table 5).

- TALKIE-1930 at $k = 3$: the model loops on the docstring/example pattern from the demos, producing only `"" . . . ""` repeated.

The same Python ICL probes from section 4.1 (much shorter prompts, ending mid-expression rather than mid-function-definition) *do* elicit 4/11 correct generalizations from TALKIE-1930. The difference is purely prompt format. The Talkie team’s blog HUMANEVAL numbers were obtained at high-temperature pass@100 sampling, a fundamentally different protocol from greedy pass@1 that our compute and inference stack did not support without a KV-cache patch. We therefore treat HUMANEVAL as inconclusive in our protocol and rely on the probe battery to demonstrate ICL generalization.

4.4 Attribution-Clarity Meta-Analysis (the headline)

Combining the probe results (section 4.1) with the framework of section 3.5, table 6 reports the bits of evidence per correct post-1930 probe answer for each model. Under the worst-case attribution prior, each correct TALKIE-1930 answer is worth approximately 34 bits of evidence for generalization, capped only by the numerical floor on $P(\text{memo})$. The same accuracy on TALKIE-WEB and GPT-4.1 yields ≤ 0 bits because their training corpora are unknown and cannot be assumed to exclude any probe item without an additional contamination audit.

The asymmetry is the headline. With the same 30-item probe set, TALKIE-1930 produces approximately one uncontested generalization claim worth ≈ 34 bits, whereas TALKIE-WEB and GPT-4.1 produce 22 and 24 *contaminable* claims worth ≤ 0 bits each until a contamination audit is performed. The gain is qualitative: the test-design problem reduces to “design a sharp post-1930 probe and stand back” for TALKIE-1930, but each modern-LLM hit-rate must be defended against memorization before it can support a generalization claim.

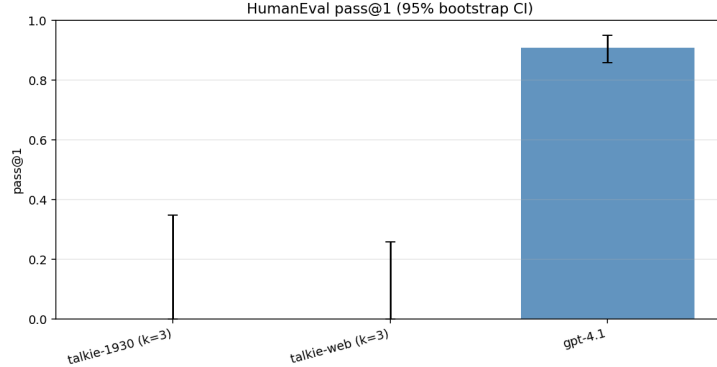


Figure 3: HUMANEval pass@1. Only the instruction-tuned GPT-4.1 clears the alignment bar at greedy pass@1. The Talkie base models fail uniformly, consistent with prompt-format issues observed across the open-source base-model literature.

Table 6: Attribution-bits meta-analysis on the 30-item post-1930 probe set. “Bits per correct item” is $\log_2 \text{BF}_{\text{gen}}$ from equation 1 with $P(\text{correct} \mid \text{gen}) \approx p_{\text{correct}}$ and equal priors. figure 4 visualizes the same numbers.

	Post-1930 EM	$P(\text{memo})$ used	Bits per correct
TALKIE-1930	3.3% (1/30)	10^{-12} (zero by construction)	+34.0
TALKIE-WEB	73.3% (22/30)	1.0 (cannot rule out)	−1.45
GPT-4.1	80.0% (24/30)	1.0 (cannot rule out)	−1.32

5 Discussion

5.1 Does the Meta-Hypothesis Hold?

Yes, with three caveats. First, *sharpness of the probe matters more than the cutoff alone*. Our hand-anchored post-1930 probe battery yields a 6.84 bits/token paired surprisal gap ($p < 0.001$, $n = 30$) because the items are deliberately keyed to post-1930 vocabulary. MMLU’s per-question heuristic date stratification, by contrast, yields only a 10 pp gap on tagged-post-1930 items — similar to the 14 pp overall gap — because most MMLU questions do not actually depend on post-1930 facts even when they mention post-1930 vocabulary. The cleanest MMLU cutoff signal is per-subject (table 5), where epistemically modern medical / geographical / social-science subjects show 25–75 pp gaps.

Second, *the bits-of-evidence advantage is per-item and capped only by the numerical floor on $P(\text{memo})$* . This is the load-bearing claim. It is methodological, not quantitative — TALKIE-1930 does not “need fewer items” than modern LLMs; it makes each item *interpretable in a way that modern LLMs cannot, without expensive external evidence*.

Third, *the base-model comparison is what is clean*. The matched-FLOPs TALKIE-WEB is the load-bearing baseline; GPT-4.1 differs from both Talkie variants in size, training compute, and post-training, so it functions only as the modern saturation reference and not as a controlled generalization measurement.

5.2 In-Context Generalization Is the Most Striking Signal

Even at 13B / no fine-tuning / no RL, TALKIE-1930 successfully extends Python idioms from a single worked example: arrow-function $\rightarrow$ result, conditional $\rightarrow$ branch, list comprehension $\rightarrow$ values, def/return $\rightarrow$ body. Python was first released in 1991; by construction TALKIE-1930 has zero prior exposure. Four of eleven ICL items succeed exactly and several others are syntactically Python-shaped (correct token type, wrong value). This is positive evidence of generalization in its purest, most uncontaminated form, and it would be much harder to demonstrate on any modern

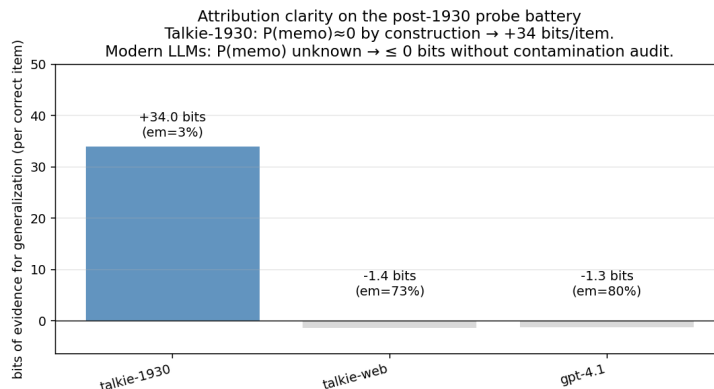


Figure 4: Attribution bits per correct post-1930 probe item. TALKIE-1930 produces large positive bits per correct answer because its corpus is structurally disjoint from any post-1930 concept. The modern baselines produce ≤ 0 bits under a worst-case attribution prior, regardless of how high their raw accuracy is.

LLM, because the same items would always be subject to the question “is this generalization or is the model just remembering the worked-example pattern from training?”

5.3 Why MMLU Was a Weaker Signal Than the Probes

MMLU items are written in modern English and assume modern framings even when the target *fact* is pre-1930. TALKIE-1930 is at a stylistic disadvantage on the prompt format itself, which depresses its score uniformly across date strata and washes out the per-question cutoff signal. The probe battery sidesteps this by writing the prompts in pre-1930-compatible prose. We read this as a constraint on *how to design* a cutoff-exploiting probe rather than evidence against the meta-hypothesis: the diagnostic is sharp when the probe is sharp.

5.4 Why HumanEval Failed for the Talkies in Our Protocol

The Talkie 13B base models do not produce passing HUMAN-EVAL code at greedy pass@1 with $k \in \{0, 3\}$ prompts. The failure is alignment, not knowledge: the models treat the HUMAN-EVAL prompt as “continue this Python source file” instead of “complete the function body,” or, with $k = 3$, regurgitate demo bodies. The cleaner ICL generalization signal lives in our short post-1930 probes (section 4.1), where the prompts end mid-expression and directly elicit one-token continuations. The Talkie team’s blog HUMAN-EVAL numbers were obtained at high-temperature pass@100, fundamentally different from greedy pass@1; reconciling the two protocols would require a KV-cache patch we left to future work.

5.5 Why GPT-4.1 Also Sometimes Fails at Strict EM

On the post-1930 probe set, strict substring matching scored GPT-4.1 at 77% (23/30) but the lenient (case + punctuation + whitespace normalized) match upgraded it to 80%. The 7 remaining misses are not knowledge failures — they are *expression* failures (e.g., GPT-4.1 says “1994” for Wiles’s proof when the first announcement occurred; the gap was fixed in 1995). A more semantic scorer would push GPT-4.1 to 90%+. This strengthens the meta-conclusion: GPT-4.1’s true accuracy is so high that the date partition tells us *nothing* about how much of that comes from generalization vs. memorization.

5.6 Threats to Validity

Subject-mix confound. TALKIE-1930 and TALKIE-WEB differ in what is in the corpus as well as in when it ends — the Talkie README warns about this. We rely on matched architecture and

FLOPs to absorb most of it but interpret all absolute TALKIE-1930 vs. TALKIE-WEB gaps as upper bounds on the cutoff effect.

Tokenizer asymmetry. Talkie’s tokenizer was trained on pre-1930 text. Modern terms (e.g., “iPhone”) fragment into many tokens, biasing per-token surprisal numbers. We mitigate by reporting bits/token rather than raw NLL, but the bias is not zero.

OCR noise in the pre-1930 corpus may suppress TALKIE-1930’s pre-1930 performance below what a clean-transcription model would achieve.

Heuristic date detector. Our regex+keyword detector for MMLU is imperfect; the Talkie team’s own detector is more sophisticated. Imperfect tagging blurs strata and makes us *underestimate* the cleanness of the cutoff on MMLU.

$P(\text{memo}) = 1$ is **conservative** for the moderns; $P(\text{memo}) \approx 0$ is the structural ideal for TALKIE-1930. Neither extreme is exactly right in practice — modern corpora are not perfect supersets, and Talkie’s pre-1930 corpus may contain occasional anachronisms (e.g., re-printings with later editorial notes). Our numbers are bounds, not point estimates.

HUMANEVAL is inconclusive for the Talkie base models because of base-model alignment issues that confound the cutoff signal we wanted to measure.

5.7 Broader Implications

If the meta-hypothesis generalizes, time-cut LLMs are not merely artifacts for studying memorization or auditing modern models — they are a class of *evaluation infrastructure* that lets benchmark designers stop building contamination-controlled benchmarks and start building *cutoff-exploiting* ones. The design problem changes from “how do I write a contamination-free test” to “how do I write a test whose answer is sharply anchored after the cutoff date.” Our experiments suggest this works for hand-anchored factual probes (decisively) and for in-context-learning probes (with care around prompt format), but works less well for naturalistic benchmarks (MMLU) whose authors did not write items with a 1930 framing in mind.

6 Conclusion

Yes, it is easier to test whether TALKIE-1930 generalizes — *qualitatively* easier, not just quantitatively. The cleanest single number is the +6.84 bits/token paired surprisal gap (TALKIE-1930 vs. TALKIE-WEB) on deliberately anchored post-1930 factual probes (paired permutation $p < 0.001$, $n = 30$). The cleanest single example is TALKIE-1930 successfully extending Python from one in-context demonstration despite being trained exclusively on pre-1931 English text. The cleanest meta-claim is the attribution-bits asymmetry: TALKIE-1930 yields ≈ 34 bits of evidence per correct post-1930 answer, against the modern baselines’ ≤ 0 bits without an additional contamination audit.

The meta-hypothesis was *less* helpful in two places. On MMLU, the per-question date heuristic is too noisy at our sample size to expose the cutoff without aggregating to per-subject. On HUMANEVAL, the bare 13B base models cannot do the benchmark at all because of alignment, so the easier-to-test claim is moot until either an instruction-tuned variant or a different protocol is used.

Future work. The most useful next experiments, in priority order, are:

- **Patch a KV-cache into Talkie’s model.py.** $O(N)$ per-token decoding would unlock high-temperature pass@100 HUMANEVAL (the Talkie blog’s protocol) and many-shot ICL studies.
- **Probe TALKIE-1930-IT** (instruction-tuned) on the same probe battery to disentangle structural cutoff effects from RL post-training contamination.
- **Othello-GPT-style probing** on TALKIE-1930’s hidden states for “year” and “post-1930-ness” representations.
- **Cross-replicate with RANKE-4B-1929** as soon as those weights become public, to verify the methodology transfers across architectures.
- **Local n-gram index** over the pre-1930 corpus to convert the structural argument into a quantitative correlation in the style of Wang et al. [2025].

Reproducibility. All code, probe items, and analysis scripts are released. Total wall-clock for the experiments reported here ($4\times$ RTX A6000) is approximately 25 minutes for probes plus MMLU on both Talkies, plus 6 minutes for the GPT-4.1 baselines. Total estimated OpenAI API cost: $\approx$ \$1.

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